IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

**COMPUTATIONAL SENSING AND IMAGING**

Monday, 18 May 2026, 10:00 AM

Time allowed: 3:00 hours

**There are ONE questions on this paper.**

**Answer ALL questions.**

**Any special instructions for invigilators and information for candidates are on page 1.**

Examiners responsible First Marker(s) : A. Bhandari
Second Marker(s) : Sen Wang

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**Department of Electronic and Electrical Engineering**

**Examination 2026**

# Computational Sensing and Imaging (ELEC70085)

**MSc and EEE/EIE PART IV**

**May, 2026 • Exam Duration: Three Hours**

**All Questions are Compulsory**

**Where unavoidable, make reasonable assumptions for your solutions.**
**Explain your assumptions in the answer script.**

**Markers: Ayush Bhandari and Sen Wang.**

**Questions [100 Marks]**

Q1 **Forward Models for Coded Illumination, Coded Aperture, and Coded Sensor.** **[25 Marks]**

In computational imaging, one can encode information at 3 different stages of the imaging pipeline:

i) *coded illumination* (before light interacts with the scene),
ii) *coded aperture* (inside the optical system),
iii) *coded sensor* (at detection, e.g., per-pixel weighting).

Consider a 1D discrete scene $\mathbf{x} \in \mathbb{R}^N$, an optical blur modeled by convolution with a kernel $\mathbf{h}$ (i.e., $\mathbf{Hx} = \mathbf{h} * \mathbf{x}$), and sensor noise $\boldsymbol{\eta}$.

(a) **Coded illumination model.** Suppose we illuminate the scene with a known illumination pattern $\mathbf{m} \in \mathbb{R}^N$ (e.g., from a projector), so that the effective scene becomes $\mathbf{m} \odot \mathbf{x}$ (pointwise multiplication). Write the forward model for the measurements $\mathbf{y}_{\text{illum}}$ in terms of $\mathbf{m}$, $\mathbf{H}$, and $\mathbf{x}$. Briefly explain (in 1-2 sentences) how coding *before the scene* affects the information captured. **[4 Marks]**

(b) **Coded aperture model.** Assume uniform illumination but insert a known mask $\mathbf{a}$ in the optical path, modifying the effective point spread function (PSF) to $\mathbf{h}_{\text{eff}} = \mathbf{a} * \mathbf{h}$. Write the forward model $\mathbf{y}_{\text{ap}}$ and explain briefly how aperture coding differs conceptually from illumination coding. **[4 Marks]**

(c) **Coded sensor model.** Finally, assume illumination and optics are standard, but the sensor applies per-pixel gains collected into a diagonal matrix $\mathbf{W} = \text{diag}(w_1, \ldots, w_N)$. Write the forward model $\mathbf{y}_{\text{sens}}$, and explain briefly what is fundamentally different about coding at the sensor compared to coding at illumination and aperture. **[4 Marks]**

(d) **Comparison.** Summarize in a simple table or 3–4 sentences the key differences among coded illumination, coded aperture, and coded sensor in terms of:

- where coding occurs,
- what part of the physics it modifies,
- and what flexibility or limitations it introduces. **[3 Marks]**

(e) **(Mix-and-match) Classifying real-world coded imaging systems (Illumination / Aperture / Sensor).** **[10 Marks]**

Below is a list of real-world imaging hardware systems discussed in the class. Each one uses coding at a specific stage of the imaging pipeline: illumination, optics, or sensor.

Match each system (1)-(10) below to exactly one of the following three categories:

I] coded illumination
A] coded aperture (coding in the optics)
S] coded sensor

1) Per-pixel coded exposure (flutter shutters on a CMOS sensor)
2) DMD-based structured light projection (e.g., binary/Gray/Hadamard patterns)
3) MURA / URA mask for high-energy (X-ray/gamma) imaging
4) Time-of-flight (ToF) sensors with per-pixel electronic modulation
5) Phase-mask wavefront coding (e.g., cubic-phase mask for extended depth-of-field)
6) Color filter arrays (e.g., Bayer, Quad-Bayer, multispectral mosaics)
7) Light-field / plenoptic camera with microlens array at an aperture conjugate plane
8) Compressive cameras with in-pixel mixing (e.g., ASPs, compressive CMOS arrays)
9) Rolling-shutter spatiotemporal coding (row-wise phase shifts)
10) Coded flash / temporally modulated illumination for motion deblurring)

**Q2** **Constrained Least Squares (CLS) for a toy imaging system.** **[20 Marks]**

Consider a simplified (toy) 2-pixel imaging system with an unknown scene vector $\mathbf{x} \in \mathbb{R}^2$ (two reflectivities). The optics/sensor pipeline is modeled as a $2 \times 2$ linear operator $\mathbf{H}$ (blur/crosstalk) followed by a per-pixel gain matrix $\mathbf{W}$ (pixel responsivity). The measurement model is

$$\mathbf{y} = \mathbf{WHx} + \boldsymbol{\eta},$$

where $\boldsymbol{\eta}$ is measurement noise.

We pose the following constrained least-squares (CLS) problem:

$$\min_{\mathbf{x}\in\mathbb{R}^2} \left\|\mathbf{WHx} - \mathbf{y}\right\|_2^2, \qquad \text{subject to} \quad \mathbf{Cx} = \mathbf{b},$$

with

$$\mathbf{W} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Interpretation: $\mathbf{Cx} = \mathbf{b}$ encodes simple scene priors/constraints (e.g., known total brightness and left-right symmetry).

(a) Write down the Lagrangian $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})$. **[4 Marks]**

(b) Derive the optimality conditions in matrix form (a linear system in $(\mathbf{x}, \boldsymbol{\lambda})$). **[6 Marks]**

(c) Solve the optimal system in closed form to obtain the optimal $\mathbf{x}^\star$. **[6 Marks]**

(d) Verify that $\mathbf{Cx}^\star = \mathbf{b}$, and report the achieved residual $\|\mathbf{WHx}^\star - \mathbf{y}\|_2^2$. **[4 Marks]**

**Q3** **One-Bit Time-Resolved Imaging.** **[15 Marks]**

*Context.* Time-resolved imaging systems (e.g., ToF cameras) attempt to measure the time-of-flight of a light pulse reflected from a scene. If a pulse $p(t)$ is emitted and the reflection is $r(t) = \Gamma p(t - t_0)$, then estimating the unknown delay $t_0$ reveals depth. However, sampling $r(t)$ at high temporal resolution (nanoseconds) across megapixel sensors lead to enormous data rates. To reduce data size, we explore combining *one-bit sampling* with time-resolved imaging.

The standard one-bit (first-order sigma–delta) scheme for a bandlimited signal $f(t)$ is:

$$\begin{cases} q[n] = \mathrm{sgn}(\, u[n-1] + f[n]\,), \\ u[n] = u[n-1] + f[n] - q[n], \end{cases}$$

where $q[n] \in \{-1, +1\}$ is the quantized output, $u[n]$ is the internal accumulator, and $\mathrm{sgn}$ is the sign or signum function.

(a) Explain why megapixel time-resolved imaging is difficult? **[3 Marks]**

(b) Let $p[n] = p(nT)$ be samples of a bandlimited pulse and let the reflected signal be

$$r[n] = \Gamma\, p(nT - t_0).$$

Using the one-bit model above, *express* $q[n]$ *using* $r[n]$ *and* $u[n]$ *alone*. **[3 Marks]**

(c) Using $q[n]$ from above and the linearity of the Discrete-Time Fourier Transform (DTFT), show that

$$\widehat{q}(\omega) \approx \Gamma e^{-j\omega t_0}\, \widehat{p}(\omega) - \widehat{u}(\omega)\, \widehat{v}(\omega),$$

for an appropriate filter $v[n]$. **[3 Marks]**

*Hint: Use* $u[n] - u[n-1] = r[n] - q[n]$*, i.e., a discrete derivative.*

(d) Our goal is to estimate $t_0$ by matched filtering. For the original (unquantized) system,

$$t_0^\star = \arg\max_t\, |(r * \overline{p})(t)|, \qquad \overline{p}(t) = p(-t).$$

Show that the *same strategy* can be applied to $q[n]$ instead of $r[n]$ to estimate $t_0$. **[6 Marks]**

*Hints:*

- Convolution with $\overline{p}$ acts as a *low-pass filter*.
- From part (c), $q = r - (u * v)$ where $v$ is a discrete derivative.
- A derivative $v$ corresponds to a high-pass filter in frequency.

**Q4** **From a Single Cosine to AR Modeling for Spectral Estimation.** **[20 Marks]**

Consider a real, noiseless discrete-time cosine

$$x[n] = c_0 \cos(\Omega_0 n + \phi_0), \qquad n = 0, \ldots, N-1,$$

with unknown amplitude $c_0 > 0$, frequency $\Omega_0 \in (0, \pi)$, and phase $\phi_0 \in (-\pi, \pi]$.

(a) Show that $x[n]$ satisfies a second-order autoregressive (AR(2)) recursion of the form

$$x[n] + a_1 x[n-1] + a_2 x[n-2] = 0, \qquad n = 2, \ldots, N-1,$$

and express $a_1, a_2$ in terms of $\Omega_0$. **[4 Marks]**

(b) Using the recursion:

$$x[n] + a_1 x[n-1] + a_2 x[n-2] = 0, \qquad n = 2, \ldots, N-1,$$

form a linear system to estimate the AR parameters $(a_1, a_2)$ from $x[n]$, and give the minimum value of $N$ required in the noiseless case. **[6 Marks]**

(c) Show how to recover $\Omega_0$ from the estimated AR coefficients, and then estimate $c_0$ and $\phi_0$ by a linear fit. **[6 Marks]**

(d) Briefly explain how this extends to $K$ distinct cosines. **[4 Marks]**

**Q5** **Sparse Signals and Smooth approximations to $\ell_1$ for Denoising** **[20 Marks]**

*Context and goal.* We observe a noisy vector $\mathbf{g} \in \mathbb{R}^N$ (e.g., wavelet coefficients of a noisy image or audio). We wish to estimate a denoised vector $\mathbf{f}$ whose entries are mostly small or zero (sparse), while staying close to the data. The classical approach uses the $\ell_1$ penalty and yields the soft-thresholding rule. Here we study a smooth approximation of the $\ell_1$ norm that is easier to optimize with gradient-based methods and to differentiate through (e.g., for learning pipelines), and we examine its shrinkage behavior and its relationship to soft-thresholding.

We consider the separable cost

$$\mathcal{C}_\lambda(\mathbf{f}) = \|\mathbf{g} - \mathbf{f}\|_2^2 \; + \; \lambda \sum_{n=1}^{N} \sqrt{f_n^2 + \varepsilon}, \qquad \lambda > 0, \quad \varepsilon > 0.$$

This replaces the non-differentiable $\ell_1$ norm by the smooth surrogate $\sqrt{f^2 + \varepsilon}$. Throughout, notice the problem decouples across coordinates; thus we can analyze the one-dimensional objective

$$\phi(f) \; = \; (g - f)^2 + \lambda \sqrt{f^2 + \varepsilon}$$

and apply the result component-wise.

(a) Explain why the surrogate penalty $\sqrt{f^2 + \varepsilon}$ is a reasonable smooth approximation to $|f|$, and why replacing $|f|$ by it can still promote sparse (or near-sparse) recovery. **[4 Marks]**
*Hint.* Sketch the graphs of $|f|$ and $\sqrt{f^2 + \varepsilon}$ for a small $\varepsilon$ and compare them near the origin.

(b) Differentiate $\phi(f)$ with respect to $f$ and derive the first-order stationarity condition for a minimizer. **[4 Marks]**

(c) Use the rearranged stationarity condition

$$g \; = \; f \left( 1 + \frac{\lambda}{2\sqrt{f^2 + \varepsilon}} \right)$$

to show that the minimizer $f^\star$ has the same sign as $g$, and that its magnitude is no larger than that of $g$. Show also that $|f^\star| < |g|$ whenever $g \neq 0$. **[4 Marks]**

(d) Show that for any finite $\varepsilon > 0$, the minimizer of $\phi(f)$ cannot be $f = 0$ whenever $g \neq 0$. **[4 Marks]**

*Hint:* Evaluate $\phi'(0)$ and recall that $\phi$ is convex.
Briefly explain what this implies about the ability of the smooth surrogate to produce exact sparsity compared with the $\ell_1$ penalty.

(e) How is this related to soft-thresholding?
Consider the limit $\varepsilon \to 0^+$ in the first-order condition and recall the soft-threshold operator.
The *soft-threshold* operator with threshold $\tau > 0$ is defined componentwise as

$$\mathrm{soft}_\tau(g) = \begin{cases} \mathrm{sgn}(g)\,(|g| - \tau), & |g| > \tau, \\ 0, & |g| \leq \tau. \end{cases}$$

Show that, in the limit $\varepsilon \to 0^+$, the minimizer approaches a soft-thresholding rule, and determine the relation between $\tau$ and $\lambda$ for the present scaling. **[4 Marks]**